ChemRMBench: A Full-Tail Cold-Continuation Audit
V99 Research Progress Addendum

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Abstract

A long-task restart can finish the task while failing its declared state-equivalence contract. We test a fresh-process continuation from an unloaded approach checkpoint in a reconstructed laboratory. The cold process executes all 11,027 remaining commands without rerunning its 2,400-command prefix, preserves the declared task clock and returns the bottle with 0.467 mm error. All seven geometric/program endpoint comparisons pass; all seven original strict comparisons fail on a saved feedback-effort target field. We preserve this negative result, identify the comparison boundary and specify a prospective all-action-channel control. This single engineering pair is not memory evaluation, chemical timing validation or a complete benchmark.

1. Cold continuation: completion and strict state agreement

Long laboratory tasks require more than a saved image or an endpoint pose. A continuation checkpoint is

$$C_k = (x_k, c_k, \ell_k, \kappa), \tag{1}$$

where x_k contains registered physical state and actuator targets, c_k is the task/controller cursor, ℓ_k is the declared task clock, and κ identifies the scene, controller, registry and original goal. Hidden solver state is not included in our current implementation. We distinguish archive integrity, initial restoration, transient holding, functional continuation and strict later-state agreement. Passing one does not establish the others.

Prospective experiment. V99 seals 132 source files and its independent reader before native startup. A fresh live reference runs all 13,427 commands and inserts a 30-step hold at the unloaded, open-gripper approach boundary, cursor 2,400. A second process reconstructs the scene, restores the saved checkpoint, performs the matched hold and executes only the remaining 11,027 commands. The first 2,400 commands are not replayed. Both conditions use the same pinned controller program, scene and local RTX 5090. At each remaining stage, the reader compares all 56 rigid bodies and nine articulations against the corresponding live checkpoint. The original limits remain 1 mm root position, 0.5° root orientation and 10^{-4} saved joint coordinate; the last quantity is expressed in each joint's stored coordinate, not uniformly in radians. The strict comparator additionally requires exact target/property records and matching host/task metadata. Initial observed velocity limits and the 30-sample hold remain separate gates.

Task clock. Let h be cold native steps since the restored boundary, including its 30 hold steps. The declared paused virtual task clock is

$$\ell_h^{\mathrm{cold}} = \ell_{2400}^{\mathrm{saved}} + h\Delta t, \qquad \Delta t \simeq 1/240\ \mathrm{s}. \tag{2}$$

The implementation uses the recorded native step duration; the saved prefix is 10.000000522 s. Fresh-scene warmup and process downtime are excluded by this explicit policy. The reader checks agreement at every remaining source command to 10^{-7} s. A video starts at presentation time zero while displaying the preserved task time. This experiment does not test continuously evolving chemistry or agent memory.

Observed result. Initial restoration and all 30 matched hold samples pass. Both conditions complete their full command sequences, with all eight live boundaries and all seven cold tail boundaries saved. The cold condition retains bilateral contact in all sampled transport steps, has maximum relative displacement 0.571 mm and returns the bottle with 0.467 mm error. This satisfies V99's prospectively used additional 1 mm completion condition; the earlier task return limit was 20 mm and is not retroactively redefined. All seven geometric/same-controller-program endpoint comparisons pass. Nevertheless, all seven original strict comparisons fail (Table 1). Thus V99 adds no verified cold boundary: coverage remains $1/8$.

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Research progress addendum. One engineering pair; not submitted or peer reviewed.

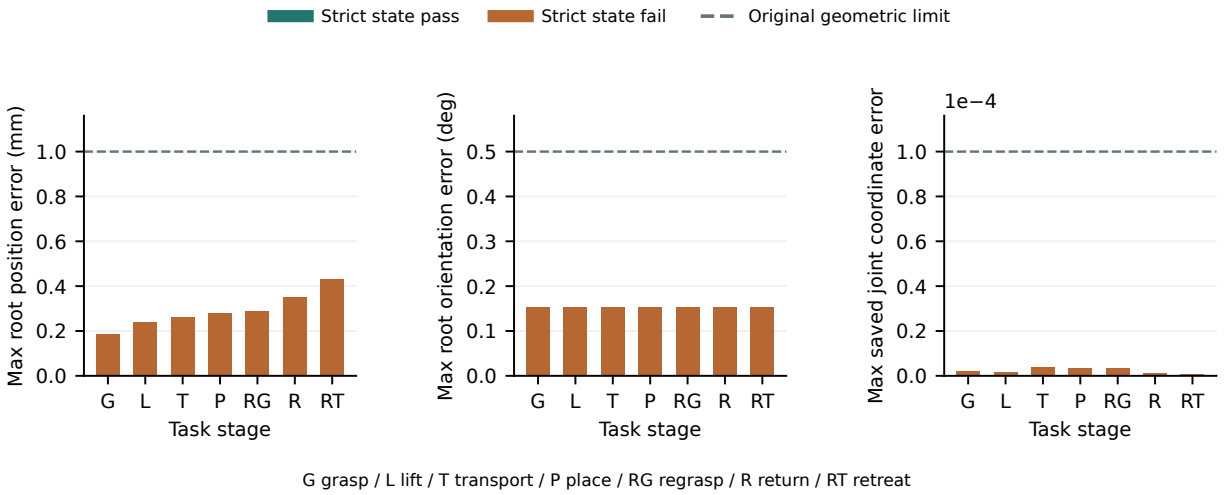


Figure 1. V99 full-tail stage endpoints. Each bar is the maximum difference over all 65 registered entities. Dashed lines retain the original geometric limits; color shows the original strict state verdict. Small geometric errors do not override differing actuator targets. The seven stages are nested within one fixed-condition live/cold pair.

Table 1. One fixed live/cold pair. Maxima cover the 65 registered entities at each endpoint. All seven strict comparisons fail; geometric/program comparisons pass. Saved joint coordinate differences are $< 4 \times 10^{-6}$. These are stages of one pair, not seven independent task trials.

Stage	Position (mm)	Angle (°)	$\max \lvert\Delta\tau\rvert$
Grasp	0.185492	0.152504	3.815×10^{-5}
Lift	0.238345	0.152504	4.196×10^{-5}
Transport	0.259081	0.152504	4.578×10^{-5}
Place/release	0.278311	0.152504	5.531×10^{-5}
Regrasp	0.286381	0.152504	3.052×10^{-5}
Return	0.349665	0.152504	5.579×10^{-5}
Retreat	0.430962	0.152504	1.717×10^{-5}

Field attribution and its limit. The only differing strict actuator field at these endpoints is `fr3_A.joint_effort_target`; host checks and registry comparisons pass. These are differences in saved effort-target coordinates, without an independently calibrated torque-unit claim. The pinned controller recomputes gravity compensation from the evolving state. In particular,

$$a_k = (u_k, g(q_k)), \qquad g(q_k^{\text{live}}) \neq g(q_k^{\text{cold}}) \tag{3}$$

can hold when u_k and the program g agree and pose errors are within the allowed tolerance. This explains why equality of a controller program does not imply equality of all future action values; it does not isolate the physical cause of state divergence. V99's sealed verdict is retained.

Next controlled experiment, not yet run. V100 will record every applied actuator channel, including the held steps and gravity-compensation efforts, then replay the same values in a fresh cold process. It keeps the original restoration and strict endpoint criteria. This separates future feedback-action mismatch from divergence under matched applied actions. The action trace is private validation data, never oracle future actions supplied to a benchmark agent. A pass would be a scoped engineering result, not memory, chemical timing, task generalization or policy-learning evidence.

Reproducibility and cost. The sealed original reader recomputes the result from raw records. The two workers close without timeout, allocating at most 0.212626 GPU h; there are no model calls or training jobs. The verified durable archive contains 1,614 files and 1,536,860,160 bytes. The native RGB movies contain 673 and 553 frames at 12 fps. A 553-frame side-by-side derivative aligns the live movie's frame 120 with the cold movie's frame zero; all paired task timestamps agree. It is not an additional physical trial.